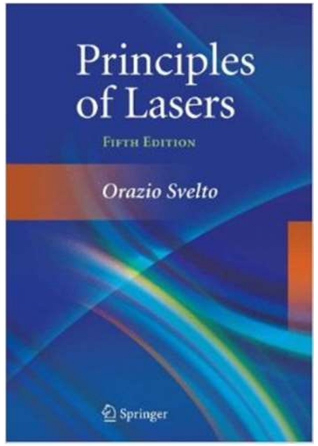
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers



Ch3-3
Resonant Interaction between Electromagnetic Field & Matter

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Chapter 3: Resonant Interaction between Electromagnetic Field & Matter

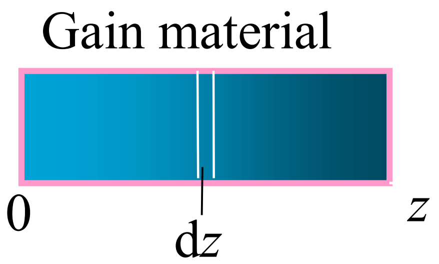
❖ 3.4 Gain coefficient (homogeneous broadening)

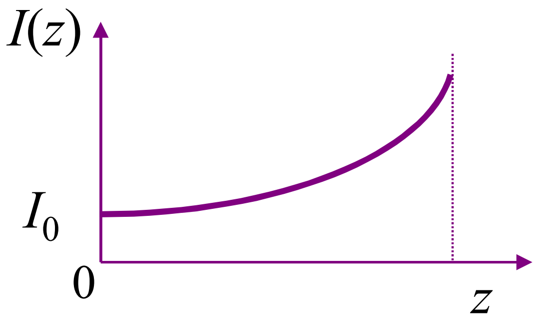
- Saturation of population inversion
- Gain saturation

❖ 3.5 Gain coefficient (inhomogeneous broadening)

- Gain saturation
- Hole-burning behavior

3.4 Gain coefficient (homogeneous)





$$\left\{\begin{aligned} \frac{dN}{dt} &= \left(n_2 - \frac{f_2}{f_1} n_1\right)\sigma_{21}(\nu,\nu_0)\upsilon N - \frac{N}{\tau_{Rl}} \quad \text{ignore loss} \\ &= \Delta n \sigma_{21}(\nu,\nu_0)\upsilon N \\ I(z) &= Nh\nu\upsilon \end{aligned}\right.$$

$$\frac{dN}{dt} = \frac{dI(z)}{h\nu\upsilon dt} = \Delta n \sigma_{21}(\nu,\nu_0)\upsilon \frac{I(z)}{h\nu\upsilon}$$

$$\Rightarrow \frac{dI}{I(z)} = \Delta n \sigma_{21}(\nu,\nu_0)\upsilon dt = \Delta n \sigma_{21}(\nu,\nu_0) dz$$

$$g(z) = \frac{dI(z)}{dz}\frac{1}{I(z)}$$

$$\Rightarrow \boxed{g = \frac{dI(z)}{dz}\frac{1}{I(z)} = \Delta n \sigma_{21}(\nu,\nu_0) = \Delta n \frac{\upsilon^2 A_{21}}{8\pi\nu_0^2}\tilde{g}(\nu,\nu_0)}$$

Gain Coefficient

$$\boxed{g = \Delta n \sigma_{21}(\nu,\nu_0) = \Delta n \frac{\upsilon^2 A_{21}}{8\pi \nu_0^2} \tilde{g}(\nu,\nu_0)}$$

❖ Saturation of population inversion (four-level system)

$$\left\{\begin{array}{l}
\dfrac{dn_3}{dt} = n_0 W_{03} - n_3\left(S_{32} + A_{30}\right) \xrightarrow{\text{equilibrium}} n_0 W_{03} \approx n_3 S_{32} \xrightarrow[S_{32} >> A_{30}]{S_{32} >> W_{03}} n_3 \approx 0 \\[2ex]
\dfrac{dn_0}{dt} = n_1 S_{10} - n_0 W_{03} + n_3 A_{30} \xrightarrow{\text{equilibrium}} n_1 S_{10} \approx n_0 W_{03} \xrightarrow{S_{10} >> W_{03}} n_1 \approx 0 \\[2ex]
\Delta n = n_2 - \dfrac{f_2}{f_1} n_1 \xrightarrow{n_1 \approx 0} \Delta n = n_2 \\[2ex]
\dfrac{dn_2}{dt} = -\left(n_2 - \dfrac{f_2}{f_1} n_1\right)\sigma_{21}(\nu,\nu_0)\upsilon N - n_2\left(S_{21} + A_{21}\right) + n_3 S_{32}
\end{array}\right.$$

$$\begin{aligned}
\frac{d\Delta n}{dt} &= -\Delta n \sigma_{21}(\nu,\nu_0)\upsilon N - \Delta n\left(S_{21} + A_{21}\right) + n_3 S_{32} \\
&= -\Delta n \sigma_{21}(\nu,\nu_0)\upsilon N - \frac{\Delta n}{\tau_2} + n_0 W_{03}
\end{aligned}$$

$$\left[\tau_2 = \frac{1}{A_{21} + S_{21}}\right]$$ lifetime of energy level E_2

$$\frac{d\Delta n}{dt} = -\Delta n \sigma_{21}(\nu,\nu_0)\upsilon N - \frac{\Delta n}{\tau_2} + n_0 W_{03} \xrightarrow[n_0=n]{\text{equilibrium}} \boxed{\Delta n = \frac{nW_{03}\tau_2}{1+\sigma_{21}(\nu_1,\nu_0)\upsilon N\tau_2}}$$

$$\boxed{\Delta n = \frac{\Delta n^0}{1+\dfrac{I_{\nu_1}}{I_s(\nu_1)}}} \quad \text{where} \quad \begin{cases} \Delta n^0 = nW_{03}\tau_2 & \text{(population inversion density for weak light)} \\ I_{\nu_1} = Nh\nu_1\upsilon & \\ I_s(\nu_1) = \dfrac{h\nu_1}{\sigma_{21}(\nu_1,\nu_0)\tau_2} \approx \dfrac{h\nu_0}{\sigma_{21}(\nu_1,\nu_0)\tau_2} & \text{(saturation intensity at frequency } \nu_1\text{)} \end{cases}$$

(1) For weak light $I_{\nu_1} << I_s(\nu_1)$ we have $\Delta n = \Delta n^0 = nW_{03}\tau_2$

(2) For intensive light we have $\Delta n < \Delta n^0$ that means $I_{\nu_1}\uparrow \Rightarrow \Delta n \downarrow$

Saturation of population inversion: when the incident light I_{v1} is comparable to $I_s(v_1)$, the decay rate of population inversion due to stimulated emission is comparable with other processes like spontaneous emission and non-radiative recombination. Therefore, when $I_{v1} << I_s(v_1)$, Δn is not related to intensity. However, when I_{v1} is comparable to $I_s(v_1)$, Δn decrease with the increase of I_{v1}. When $I_{v1}=I_s(v_1)$, $\Delta n= \Delta n^0/2$

$I_s(v_1)$ is determined by the property of gain materials and frequency of incident light. It can be measured (as shown in Appendix I)

Saturated intensity $I_s(\nu_1)$ is inversely proportional to the line function. For homogenous broadening, we know that the line function has Lorentz shape

$$I_s(\nu_1) = \frac{h\nu_0}{\sigma_{21}(\nu_1,\nu_0)\tau_2} \qquad \sigma_{21}(\nu_1,\nu_0) = \frac{A_{21}\upsilon^2}{8\pi\nu_0^2}\tilde{g}(\nu_1,\nu_0)$$

$$\tilde{g}_H(\nu_1,\nu_0) = \frac{\Delta\nu_H/2\pi}{(\nu_1-\nu_0)^2+(\Delta\nu_H/2)^2}$$

$$\frac{I_s(\nu_1)}{I_s} = \frac{I_s(\nu_1)}{\dfrac{h\nu_0}{\sigma_{21}\tau_2}} = 1+\frac{4(\nu_1-\nu_0)^2}{(\Delta\nu_H)^2}$$

$$I_s = \frac{h\nu_0}{\sigma_{21}\tau_2}$$

saturated intensity at central frequency ν_0

Saturated intensity has minimum value at $\nu_1=\nu_0$, and it equal to the saturated intensity at central frequency

Characteristics of the saturation of population inversion

1. The saturation intensity has minimum value at central frequency, which is equal to $I_s = h\nu_0/\sigma_{21}\tau_2$. When the frequency of the incident light is far away from the central frequency, the corresponding saturation intensity will increase.

2. For specific incident light intensity: the smaller saturation intensity, the quicker decrease of Δn, and therefore the more serious saturation effect.

3. The saturation effect is most pronounced when the incident light frequency matches the central frequency. As the frequency deviates further from the central frequency, the saturation effect weakens.

WHY?

Stimulated emission is most likely to occur at the central frequency. The incident light causes a rapid decrease in the number of inverted carriers.

4. The influence of incident light with different frequencies on the population inversion is not the same.

$$\Delta n=\frac{\Delta n^0}{1+\frac{I_{\nu_1}}{I_s(\nu_1)}} \quad \xrightarrow{\text{Lorentz shape}} \quad \Delta n=\frac{(\nu_1-\nu_0)^2+(\Delta\nu_H/2)^2}{(\nu_1-\nu_0)^2+\left(\frac{\Delta\nu_H}{2}\right)^2\left(1+\frac{I_{\nu_1}}{I_s}\right)}\Delta n^0$$

Only when $|\nu_1-\nu_0|<\sqrt{1+\frac{I_{\nu_1}}{I_s}}\frac{\Delta\nu_H}{2}$, there is significant saturation effect

Note: What we discussed herein is the four-level system. For other systems, the gain coefficient has similar expression. However, the expression of the saturation intensity is not the same.

❖ Gain saturation

Gain saturation: the gain coefficient decrease with the increase of light intensity

1. The expression of the gain coefficient when incident light with frequency ν_1 and intensity $I_{\nu 1}$ passes through a homogenous gain medium

For monochromatic light with frequency ν_1 and light intensity $I_{\nu 1}$, what is the gain coefficient $g_H(\nu_1, I_{\nu 1})$ when the light passes through a gain material with homogeneous broadening?

$$\left.\begin{aligned} & g_H\left(\nu_1, I_{\nu_1}\right) = \Delta n \sigma_{21}\left(\nu_1, \nu_0\right) \\ & \Delta n = \frac{\Delta n^0}{1 + \dfrac{I_{\nu_1}}{I_s\left(\nu_1\right)}} \end{aligned}\right\} \quad g_H\left(\nu_1, I_{\nu_1}\right) = \frac{\Delta n^0 \sigma_{21}\left(\nu_1, \nu_0\right)}{1 + \dfrac{I_{\nu_1}}{I_s\left(\nu_1\right)}} = \frac{g_H^0\left(\nu_1\right)}{1 + \dfrac{I_{\nu_1}}{I_s\left(\nu_1\right)}}$$

Characteristics of the gain saturation

1. For weak incident light $I_{\nu_1} << I_s(\nu_1)$, the gain coefficient is not related to the incident light intensity. The gain coefficient is equal to $g_H^0(\nu_1)$

2. When I_{ν_1} is comparable to $I_s(\nu_1)$, the value of $g_H(\nu_1, I_{\nu_1})$ will decrease with the increase of $I_s(\nu_1)$, this is the gain saturation effect.

3. When $I_{\nu_1} = I_s(\nu_1)$, we have $g_H(\nu_1, I_{\nu_1}) = g_H^0(\nu_1)/2$

4. For specific incident light intensity: if the frequency of the incident light is equal to the central frequency, the gain saturation effect is the most serious one. While the frequency is far away from the central frequency, the gain saturation effect is weak.

$$g_H^0(\nu_1) = \Delta n^0 \sigma_{21}(\nu_1,\nu_0) = \Delta n^0 \frac{\upsilon^2}{8\pi\nu_0^2} A_{21}\tilde{g}(\nu_1,\nu_0)$$

The gain coefficient for weak incident light is not related to the frequency of incident light. Its shape is totally determined by the line function $\tilde{g}(\nu_1,\nu_0)$

$$g_H(\nu_1, I_{\nu_1}) = \Delta n \sigma_{21}(\nu_1,\nu_0) = \Delta n \frac{\upsilon^2}{8\pi\nu_0^2} A_{21}\tilde{g}(\nu_1,\nu_0)$$

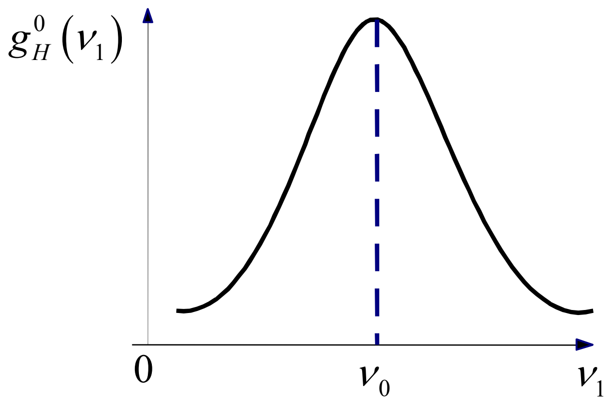
$$= \frac{(\nu_1-\nu_0)^2 + (\Delta\nu_H/2)^2}{(\nu_1-\nu_0)^2 + \left(\frac{\Delta\nu_H}{2}\right)^2\left(1+\frac{I_{\nu_1}}{I_s}\right)} \Delta n^0 \frac{\upsilon^2}{8\pi\nu_0^2} A_{21} \frac{\Delta\nu_H/2\pi}{(\nu_1-\nu_0)^2 + (\Delta\nu_H/2)^2}$$

$$= \Delta n^0 \frac{A_{21}\upsilon^2}{8\pi\nu_0^2} \frac{2}{\pi\Delta\nu_H} \frac{(\Delta\nu_H/2)^2}{(\nu_1-\nu_0)^2 + \left(\frac{\Delta\nu_H}{2}\right)^2\left(1+\frac{I_{\nu_1}}{I_s}\right)}$$

$$= \Delta n^0 \sigma_{21} \frac{(\Delta\nu_H/2)^2}{(\nu_1-\nu_0)^2 + \left(\frac{\Delta\nu_H}{2}\right)^2\left(1+\frac{I_{\nu_1}}{I_s}\right)} = g_H^0(\nu_0) \frac{(\Delta\nu_H/2)^2}{(\nu_1-\nu_0)^2 + \left(\frac{\Delta\nu_H}{2}\right)^2\left(1+\frac{I_{\nu_1}}{I_s}\right)}$$

(1) The gain coefficient $g_H^0(\nu_1)$ for weak incident light is not related to light intensity and can be expressed as

$$g_H^0(\nu_1) = g_H^0(\nu_0)\frac{(\Delta\nu_H/2)^2}{(\nu_1-\nu_0)^2+\left(\frac{\Delta\nu_H}{2}\right)^2}$$



gain coefficient for weak incident light

(2) The gain coefficient at central frequency for weak incident light

$$g_H^0(\nu_0) = \Delta n^0\sigma_{21} = \Delta n^0\frac{A_{21}\upsilon^2}{8\pi\nu_0^2}\frac{2}{\pi\Delta\nu_H} = \Delta n^0\frac{A_{21}\upsilon^2}{4\pi^2\nu_0^2\Delta\nu_H} \qquad \Delta n^0 = nW_{03}\tau_2$$

$g_H^0(\nu_0)$ is determined by the properties of gain materials and the pumping speed. It can be measured experimentally.

❖ Gain saturation

2. The expression of the gain coefficient of weak light (with frequency v) when the incident light (with frequency v_1 and intensity I_{v1}) is intensive.

For monochromatic light with frequency v_1 and strong intensity I_{v1}, along with weak light at frequency v. What is the gain coefficient of the weak light $g_H(v, I_{v1})$?

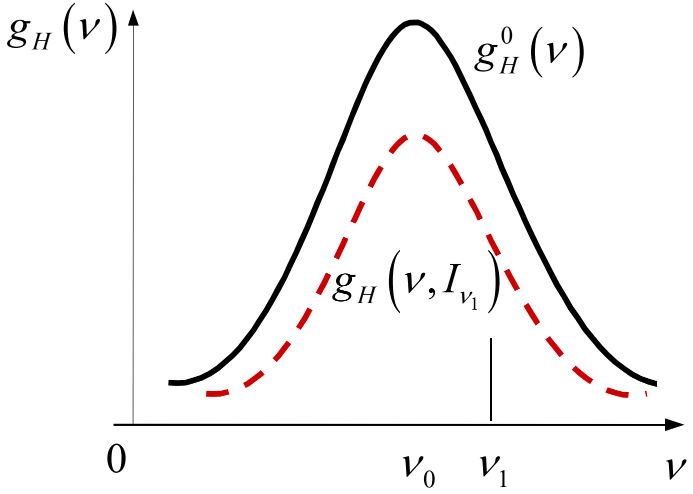
For gain materials with homogeneous broadening, the intensive light will decrease the population inversion density Δn, and the decrease of Δn will result in the decrease of gain coefficient.

$$g_H\left(\nu, I_{\nu_1}\right) = \Delta n \sigma_{21}\left(\nu, \nu_0\right) = \frac{\Delta n^0 \sigma_{21}\left(\nu, \nu_0\right)}{1 + \dfrac{I_{\nu_1}}{I_s\left(\nu_1\right)}} = \frac{g_H^0\left(\nu\right)}{1 + \dfrac{I_{\nu_1}}{I_s\left(\nu_1\right)}}$$

$$g_H\left(\nu_1, I_{\nu_1}\right) = \frac{g_H^0\left(\nu_1\right)}{1 + \frac{I_{\nu_1}}{I_s\left(\nu_1\right)}} \qquad g_H\left(\nu, I_{\nu_1}\right) = \frac{g_H^0\left(\nu\right)}{1 + \frac{I_{\nu_1}}{I_s\left(\nu_1\right)}}$$

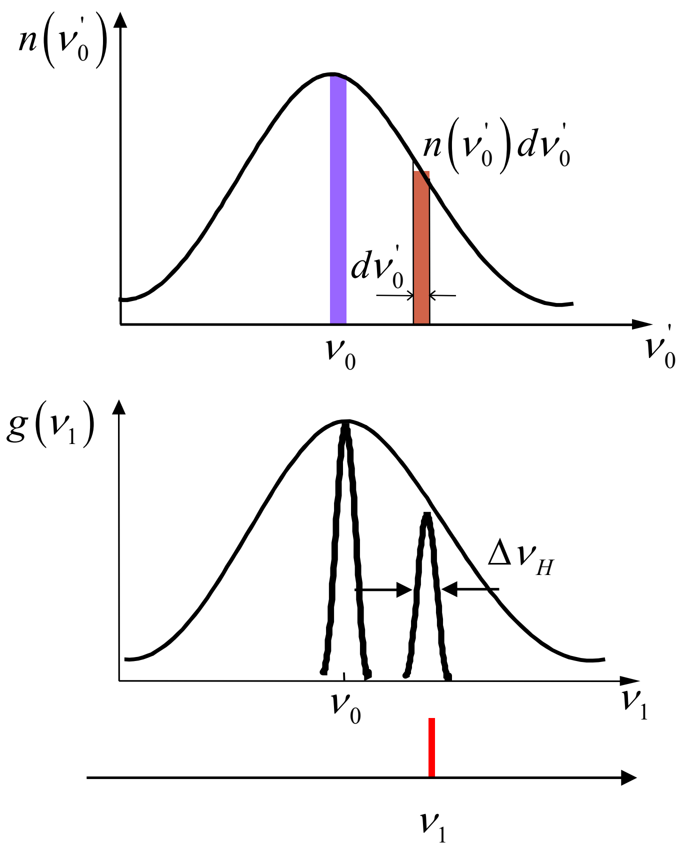
The intensive light with frequency of ν_1 not only lead to the decrease of its gain coefficient $g_H(\nu_1, I_{\nu 1})$, but also lead to the decrease of other weak light (with frequency ν) with the same degree of $g_H(\nu, I_{\nu 1})$.

For homogeneous broadening, every particles contribute to the gain spectrum at different frequencies. Therefore, once the stimulated emission with frequency of ν_1 consume the inverted carriers, it will lead to the decrease of inverted carriers with other frequency ν. Then the whole gain spectrum decrease homogeneously. In homogeneous lasers, when one mode begins to oscillate, the gain for other modes decreases, thereby preventing their oscillation (mode competition).



3.5 Gain coefficient (inhomogeneous)

- Gain saturation



$$g_H(\nu_1, I_{\nu_1}) = \frac{g_H^0(\nu_0')\left(\frac{\Delta\nu_H}{2}\right)^2}{(\nu_1-\nu_0)^2+\left(\frac{\Delta\nu_H}{2}\right)^2\left(1+\frac{I_{\nu_1}}{I_s}\right)}$$

$$g_H^0(\nu_0') = \Delta n^0(\nu_0')\frac{\upsilon^2 A_{21}}{4\pi^2\nu_0^2\Delta\nu_H}$$

$$\Delta n^0(\nu_0')d\nu_0' = \Delta n^0\tilde{g}_i(\nu_0',\nu_0)d\nu_0'$$

$$dg \approx \frac{\upsilon^2 A_{21}\overbrace{\left[\Delta n^0\tilde{g}_i(\nu_0',\nu_0)d\nu_0'\right]}^{\Delta n^0(\nu_0')d\nu_0'}(\Delta\nu_H/2)^2}{4\pi\nu_0^2\Delta\nu_H\left[(\nu_1-\nu_0')^2+\left(\frac{\Delta\nu_H}{2}\right)^2\left(1+\frac{I_{\nu_1}}{I_s}\right)\right]}$$

$$dg \approx \frac{\upsilon^2 A_{21}\left[\Delta n^0 \tilde{g}_i\left(\nu_0', \nu_0\right) d\nu_0'\right]\left(\frac{\Delta\nu_H}{2}\right)^2}{4\pi\nu_0^2\Delta\nu_H\left[\left(\nu_1-\nu_0'\right)^2+\left(\frac{\Delta\nu_H}{2}\right)^2\left(1+\frac{I_{\nu_1}}{I_s}\right)\right]}$$

$$g\left(\nu_1, I_{\nu_1}\right)=\int dg$$

$$g_i\left(\nu_1, I_{\nu_1}\right)=\frac{\upsilon^2 A_{21}\Delta n^0}{4\pi^2\nu_0^2\Delta\nu_H}\left(\frac{\Delta\nu_H}{2}\right)^2\int_0^\infty \frac{\tilde{g}_i\left(\nu_0', \nu_0\right) d\nu_0'}{\left(\nu_1-\nu_0'\right)^2+\left(\frac{\Delta\nu_H}{2}\right)^2\left(1+\frac{I_{\nu_1}}{I_s}\right)}$$

$$\Delta\nu_i >> \Delta\nu_H \qquad \tilde{g}_i\left(\nu_0', \nu_0\right) \approx \tilde{g}_i\left(\nu_1, \nu_0\right) \qquad \left|\nu_1-\nu_0'\right| < \Delta\nu_H/2$$

$$g_i\left(\nu_1, I_{\nu_1}\right)=\frac{\upsilon^2 A_{21}\Delta n^0}{4\pi^2\nu_0^2\Delta\nu_H}\left(\frac{\Delta\nu_H}{2}\right)^2\tilde{g}_i\left(\nu_1, \nu_0\right)\int_{-\infty}^{+\infty} \frac{d\nu_0'}{\left(\nu_1-\nu_0'\right)^2+\left(\frac{\Delta\nu_H}{2}\right)^2\left(1+\frac{I_{\nu_1}}{I_s}\right)}$$

$$g_i\left(\nu_1, I_{\nu_1}\right) = \frac{\upsilon^2 A_{21}\Delta n^0}{4\pi^2\nu_0^2\Delta\nu_H}\left(\frac{\Delta\nu_H}{2}\right)^2 \tilde{g}_i\left(\nu_1,\nu_0\right)\int_{-\infty}^{+\infty}\frac{d\nu_0'}{\left(\nu_1-\nu_0'\right)^2+\left(\frac{\Delta\nu_H}{2}\right)^2\left(1+\frac{I_{\nu_1}}{I_s}\right)}$$

$$= \frac{\upsilon^2 A_{21}\Delta n^0}{8\pi\nu_0^2\sqrt{\left(1+\frac{I_{\nu_1}}{I_s}\right)}}\tilde{g}_i\left(\nu_1,\nu_0\right)$$

$$\int\frac{1}{a^2+u^2}du = \frac{1}{a}arctg\frac{u}{a}$$

$$= g_i^0\left(\nu_1\right)$$

(1) For weak light $I_{\nu_1} << I_s$ we have $g_i^0\left(\nu_1\right) = \frac{\upsilon^2 A_{21}\Delta n^0}{8\pi\nu_0^2}\tilde{g}_i\left(\nu_1,\nu_0\right) = \Delta n^0\sigma_{21}\left(\nu_1,\nu_0\right)$

(2) For intensive light we have $g_i\left(\nu_1, I_{\nu_1}\right) < g_i^0\left(\nu_1\right)$ that means $I_{\nu_1}\uparrow\Rightarrow g_i\left(\nu_1, I_{\nu_1}\right)\downarrow$

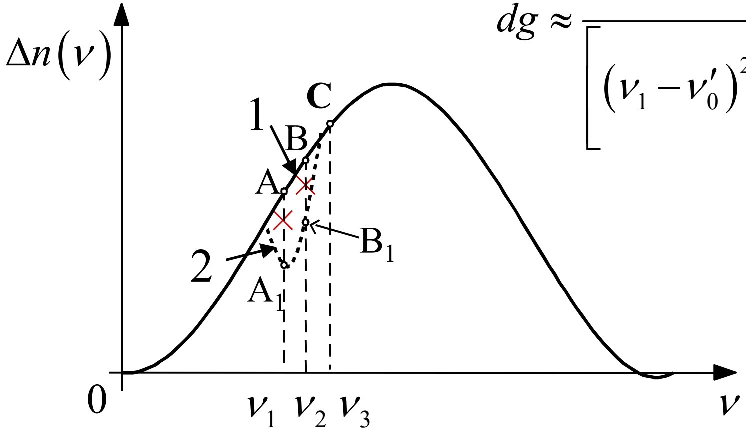
(3) For Doppler inhomogeneous broadening $\tilde{g}_i\left(\nu_1,\nu_0\right) = \tilde{g}_D\left(\nu_1,\nu_0\right)$

$$g_i^0\left(\nu_1\right) = g_i^0\left(\nu_0\right)e^{-4\ln 2\left(\frac{\nu_1-\nu_0}{\Delta\nu_D}\right)^2} \quad \text{where} \quad g_i^0\left(\nu_0\right) = \Delta n^0\frac{\upsilon^2 A_{21}}{4\pi\nu_0^2\Delta\nu_D}\left(\frac{\ln 2}{\pi}\right)^{1/2}$$

The gain coefficient of intensive light for Doppler inhomogeneous broadening medium

$$\boxed{g_i\left(\nu_1, I_{\nu_1}\right) = \frac{g_i^0\left(\nu_1\right)}{\sqrt{\left(1+\frac{I_{\nu_1}}{I_s}\right)}} = \frac{g_i^0\left(\nu_0\right)}{\sqrt{\left(1+\frac{I_{\nu_1}}{I_s}\right)}}\exp\left(-\frac{4\ln 2\left(\nu_1-\nu_0\right)^2}{\Delta\nu_D^2}\right)}$$

❖ Spectral hole-burning behavior



$$dg \approx \frac{1}{\left[\left(\nu_1 - \nu_0'\right)^2 + \left(\frac{\Delta\nu_H}{2}\right)^2 \left(1 + \frac{I_{\nu_1}}{I_s}\right)\right]}$$

$$\Delta n\left(\nu_1\right) = \frac{\Delta n^0\left(\nu_1\right)}{1 + \frac{I_{\nu_1}}{I_s}}$$

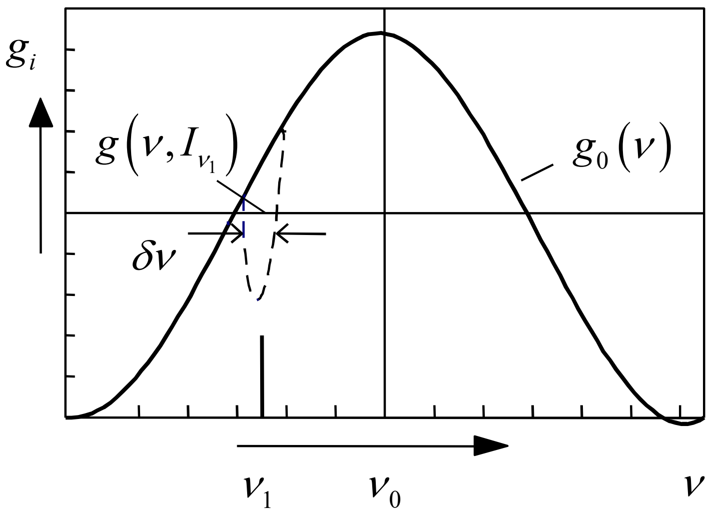
Point A: $\nu = \nu_1 \qquad A \Rightarrow A_1$

Point B: $\nu = \nu_2 \qquad B \Rightarrow B_1$

Point C: $\nu_3 - \nu_1 > \sqrt{1 + \frac{I_{\nu_1}}{I_s}}\,\frac{\Delta\nu_H}{2}$

$$\Delta n\left(\nu_3\right) = \Delta n^0\left(\nu_3\right)$$

Hole-burning behavior of gain spectrum



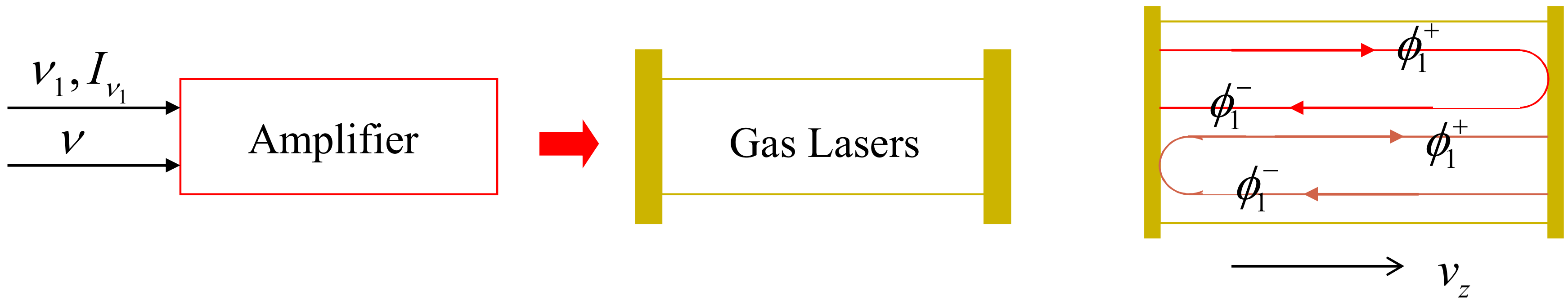
Only effective for $\nu - \nu_1 \le \pm\sqrt{1 + \frac{I_{\nu_1}}{I_s}}\,\frac{\Delta\nu_H}{2}$

depth $\quad \Delta n^0\left(\nu_1\right) - \Delta n\left(\nu_1\right) = \frac{I_{\nu_1}}{I_{\nu_1} + I_s}\Delta n^0\left(\nu_1\right)$

width $\quad \delta\nu = \sqrt{1 + \frac{I_{\nu_1}}{I_s}}\,\Delta\nu_H$

area $\quad \delta s \approx \Delta n^0\left(\nu_1\right)\Delta\nu_H \frac{I_{\nu_1}/I_s}{\sqrt{1 + \frac{I_{\nu_1}}{I_s}}}$

For Doppler inhomogeneous broadening



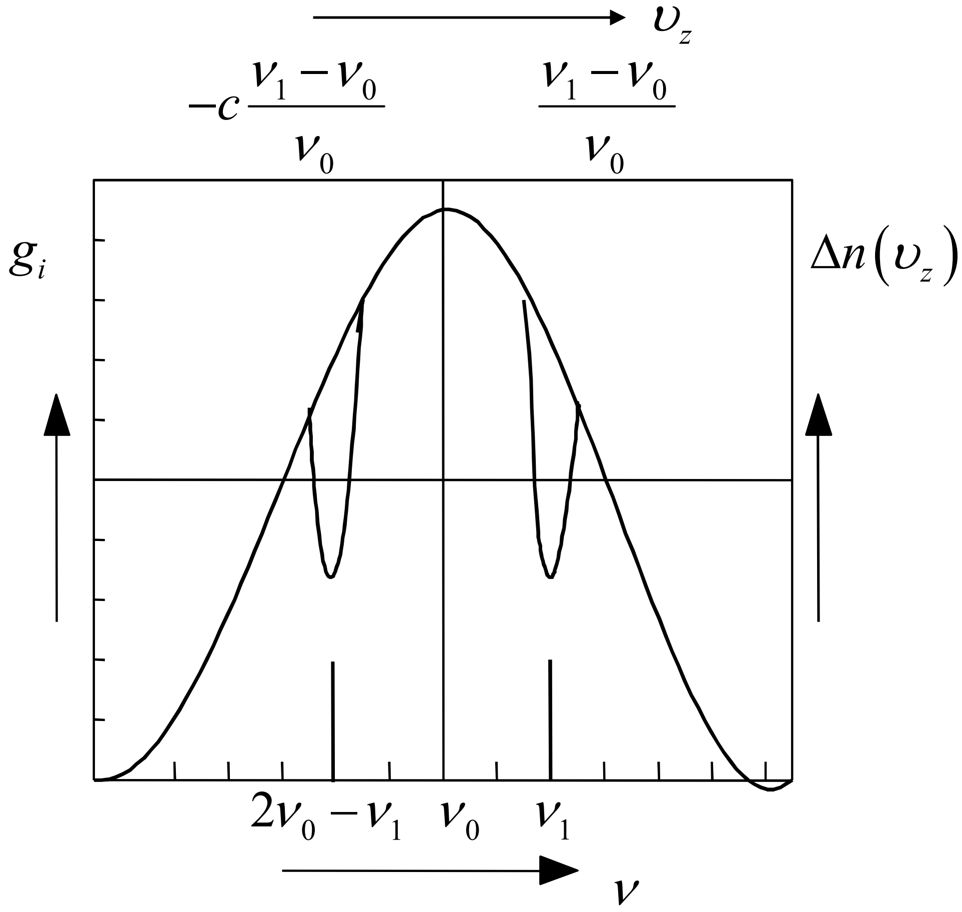
Intensive ν_1 , $I_{\nu_1} \sim I_s$ $\rightleftarrows$ Weak ν , $I_\nu \ll I_s$ $\rightleftarrows$ Gain coefficient

$\nu_1 \quad \phi_1^+ \longrightarrow$ Interact with atom ν_0, υ_z $\quad \nu_0' \approx \nu_0\left(1+\dfrac{\upsilon_z}{c}\right) \xrightarrow{\nu_1 = \nu_0'}$ $\boxed{\upsilon_z = c\dfrac{\nu_1 - \nu_0}{\nu_0}}$

ϕ_1^+ lead to the stimulated emission of v_z particle, that is

$\nu_1 \quad \phi_1^- \longleftarrow$ Interact with atom ν_0, υ_z $\quad \nu_0' \approx \nu_0\left(1-\dfrac{\upsilon_z}{c}\right) \xrightarrow{\nu_1 = \nu_0'}$ $\boxed{\upsilon_z = -c\dfrac{\nu_1 - \nu_0}{\nu_0}}$

ϕ_1^- lead to the stimulated emission of v_z particle, that is



For Doppler inhomogeneous broadening such as gas laser, the oscillated mode with frequency of v_1 will lead to two holes at the curve of gain coefficient. This two holes are symmetrically distribute on the two sides of the central frequency.